# Integration

Lecture and worked examples

# Objectives

Calculate and bound an integral; distinguish area from integral; use integration by parts; study a sequence of integrals.

# Area and antiderivative

For $f$ continuous and non-negative on $[a, b]$, $a \leq b$, the integral $\int_a^b f(x)\,\mathrm{d}x$ is the area under the curve, above the x-axis, between $x = a$ and $x = b$, expressed in area units.

# Area and antiderivative

If $f$ is continuous on an interval $I$ and $a \in I$, then $F_a(x) = \int_a^x f(t)\,\mathrm{d}t$ is the antiderivative of $f$ that vanishes at $a$. For any antiderivative $F$,

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Every continuous function on an interval has antiderivatives. An integral is a number; an antiderivative is a function.

# Area and antiderivative

For $h > 0$, the rectangles give $hf(x) \leq F_a(x+h) - F_a(x) \leq hf(x+h)$. After dividing by $h$ and taking the limit, the rate of change tends to $f(x)$. For $h < 0$, the function being increasing gives $f(x+h) \leq f(t) \leq f(x)$ on $[x+h, x]$. Integrating,

$$(-h)f(x+h) \leq F_a(x) - F_a(x+h) \leq (-h)f(x).$$

Dividing by $-h > 0$, we obtain

$$f(x+h) \leq \frac{F_a(x+h) - F_a(x)}{h} \leq f(x).$$

Both bounds tend to $f(x)$. By continuity, $F_a' = f$. Any other antiderivative is $F_a + C$, hence the formula for evaluating between the limits.

# Properties

For continuous functions and limits belonging to the same interval:

$$\int_a^a f = 0, \quad \int_b^a f = -\int_a^b f, \quad \int_a^c f = \int_a^b f + \int_b^c f, \quad \int_a^b (\lambda f + \mu g) = \lambda \int_a^b f + \mu \int_a^b g.$$

If $a \leq b$ and $f \leq g$, then $\int_a^b f \leq \int_a^b g$. In particular, if $m \leq f \leq M$, then $m(b-a) \leq \int_a^b f \leq M(b-a)$. For a function of arbitrary sign, the integral is defined using antiderivatives. The geometric area between the curve and the axis is $\int_a^b |f|$, obtained by splitting at the sign changes. Between two curves, it equals $\int_a^b |f-g|$.

# Properties

$\int_{-1}^{1} x \, \mathrm{d}x = 0$, but the area between the line and the axis over this interval is 1.

# Mean value and integration by parts

For $a < b$, the mean value is $\mu = \dfrac{1}{b-a} \int_a^b f(x)\, \mathrm{d}x$. The rectangle of height $\mu$ and width $b - a$ has the same signed area.

# Mean value and integration by parts

If $u, v$ are of class $\mathcal{C}^1$ on $[a, b]$,

$$\int_a^b u(x)v'(x)\ \mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\ \mathrm{d}x.$$

# Mean value and integration by parts

Integrate $(uv)' = u'v + uv'$ then isolate the required integral. The bracket is indispensable.

# Mean value and integration by parts

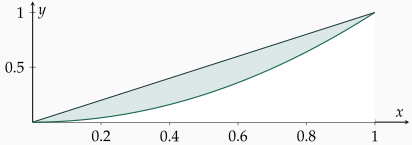
With $u = x$ and $v' = e^x$, $\int_0^1 xe^x \, \mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x \, \mathrm{d}x = e - (e-1) = 1$.

# Rectangles and sequences of integrals

For $f$ continuous and increasing on $[a, b]$, with $a < b$ and $n \geq 1$ and $h = \dfrac{b-a}{n}$, the sums $h\sum_{k=0}^{n-1} f(a+kh)$ and $h\sum_{k=1}^{n} f(a+kh)$ bound the integral. Their difference is $h(f(b) - f(a))$. For $I_n = \int_0^1 x^n f(x)\,\mathrm{d}x$, begin with the sign and comparisons of the integrand. If $0 \leq f \leq M$, then $0 \leq I_n \leq \dfrac{M}{n+1}$; the squeeze theorem gives $I_n \to 0$. Integration by parts can also provide a relation between $I_n$ and $I_{n+1}$.

# Rectangles and sequences of integrals

In $I_n = \int_0^1 x^n f(x)\ \mathrm{d}x$, the expression $x^0$ denotes the constant function 1, including at the endpoint 0. Thus $I_0 = \int_0^1 f(x)\ \mathrm{d}x$; no indeterminate calculation is needed. The area between $y = x$ and $y = x^2$ on $[0, 1]$ equals $\dfrac{1}{6}$.



**Bounding an integral with rectangles - 35 min.**

We wish to bound $I = \int_0^1 x^2 \, \mathrm{d}x$. For an integer $n \geq 1$, we split $[0; 1]$ into $n$ intervals of equal length. Let $L_n$ be the sum of the areas of the rectangles whose heights are the values of $x^2$ at the left endpoints, and $U_n$ the sum obtained at the right endpoints.

# Research activity (2)

1. For $n = 4$, draw both families of rectangles, then calculate $L_4$ and $U_4$.
2. Justify that $L_4 \leq I \leq U_4$.
3. Write a Python function with argument $n$ that returns $L_n$ and $U_n$. Determine $U_n - L_n$ algebraically.
4. Find the smallest integer $n$ for which this bounding has a width at most equal to 0.01.
5. Calculate the exact value of $I$ using an antiderivative and check that it lies within the bounds obtained for $n = 4$.